

Online Appendix

Sourcing under Sanctions: Judicial Urgency and Pharmaceutical Procurement Costs

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Supplementary material for the short paper. All empirical values are generated by the v9 build pipeline and imported through `values.tex`, generated tables, or generated figures.

A. Data Construction and Classifier Validation

The procurement data come from BEC pharmaceutical purchases in group 65 during 2009–2019 and are linked to litigation and administrative-process markers. The classifier operates at the purchase-order/tender-notice level. The empirical analysis is conducted at the purchase-offer-item level after classified regimes are linked to BEC item-level records and the relevant sample restrictions are applied. Price regressions use accepted winning bids.

The classification procedure combines machine-learning predictions, structured JSON flags, regex patterns, and position-based checks in tender notices. The production rule uses weights 3, 3, 1, and 1 for these signals and classifies a purchase regime when the vote reaches 2. Boilerplate language is filtered using 10 exclusion patterns.

Table A.1: Purchase-regime classifier validation.

Validation object	<i>N</i>	Agree.	Prec.	Recall	F1	Source
Exact agreement	179,148	98.6%	–	–	–	Ground truth
Judicial class	–	–	0.87	1.00	0.93	Ground truth
Administrative class	–	–	0.92	0.99	0.96	Ground truth
Urgent-class macro-F1	–	–	–	–	0.94	Urgent classes
Judicial class	34,191	–	0.88	0.99	0.93	ML hold-out
Administrative class	34,191	–	0.91	0.98	0.94	ML hold-out

Notes: The classifier operates at the purchase-order/tender-notice level. The table is generated from the production classification report and classifier code. Exact agreement counts 176,713 matched ground-truth labels. The empirical POI-level data link these classified regimes to item-level procurement records.

The classifier universe contains 764,362 classified purchase orders/tender notices. The full BEC pharmaceutical analysis file contains 479,330 purchase-offer-item observations. This unit distinction

Table A.2: Classifier confusion counts for urgent regimes.

Class	True positive	False positive	False negative	True negative
Judicial	11,362	1,746	46	165,994
Administrative	8,492	698	49	169,909

Notes: Counts are one-versus-rest validation counts at the purchase-order/tender-notice level. Because a purchase order can contain both judicial and administrative markers in the ground-truth file, the two rows are separate binary validations rather than mutually exclusive rows of a single multinomial matrix.

explains why the classifier universe is larger than the POI-level empirical sample. Validation against 179,148 ground-truth purchase orders gives 98.6% exact agreement. The urgent-class F1 scores are 0.93 for judicial purchases and 0.96 for administrative purchases; the urgent-class macro-F1 is 0.94. Remaining false positives and false negatives would attenuate regime contrasts if they are not systematically correlated with residual prices within controls.

Table A.3: Classifier-error sensitivity diagnostics.

Class	FPR	FNR	Contamination	Attenuation
Judicial	1.04%	0.40%	13.3%	0.986
Administrative	0.41%	0.57%	7.6%	0.990

Notes: The false-positive rate is false positives divided by false positives plus true negatives in the purchase-order validation file. The false-negative rate is false negatives divided by true positives plus false negatives. Predicted-label contamination is one minus precision. The attenuation factor is the one-versus-rest diagnostic $1 - \text{FPR} - \text{FNR}$; it is not used as a structural correction to price estimates because the validation file is at the purchase-order/tender-notice level, while the price regressions are POI-level.

Table A.3 reports a classifier-error sensitivity diagnostic at the purchase-order validation level. The diagnostic rates are small relative to the regime differences studied in the paper and mainly support an attenuation interpretation. The table is not a substitute for the mechanism evidence because validation labels do not carry POI-level prices.

B. Sample Construction and Balance

The empirical samples differ by unit and restriction. The classifier universe is the purchase-order/tender-notice file. The full BEC pharmaceutical sample contains 479,330 purchase-offer-item observations. The analysis sample contains 226,305 POI observations after the core item and variable restrictions. The winners-only price sample contains 196,883 accepted winning bids. The urgent panel contains 56,803 winner observations used for the administrative-versus-litigated

comparison. The firm-buyer-item triple sample contains 4,573 observations in 1,206 triples and is used only to isolate same-firm pricing.

Table B.1: Sample construction and empirical units.

Object or sample	Size	Unit and use
Classifier universe	764,362	Purchase orders/tender notices classified into procurement regimes.
Full BEC pharmaceutical file	479,330	Purchase-offer-item observations after linking BEC item records to regime classifications.
Analysis sample	226,305	POI observations satisfying core item and variable restrictions.
Winners-only price sample	196,883	Accepted winning bids used in negotiated-price regressions.
Urgent panel	56,803	Administrative and litigated urgent winning bids used for the under-the-gun comparison.
Both-regime urgent cells	5,581	Item-by-year-month cells containing both administrative and litigated urgent purchases.
Singleton urgent cells	11,972	Comparable urgent cells containing only one urgent regime.
Firm-buyer-item triples	1,206 triples; 4,573 observations	Same supplier, buyer, and item comparisons used to isolate within-firm pricing.

Notes: This table separates the classifier unit from the empirical regression units. POI denotes purchase-offer-item. Price regressions use accepted winning bids, while classifier validation operates at the purchase-order/tender-notice level.

Administrative and litigated urgent purchases are not balanced by design. Administrative purchases are selected, larger, and differ in item composition. The design therefore combines selection bounds, within-firm pricing tests, and direct sourcing evidence rather than treating the administrative channel as randomly assigned.

Table B.2: Sample-flow diagnostics for POI-level analysis samples.

Sample	Unit	Size	Parent sample	Share of parent
Full BEC pharmaceutical file	POI	479,330	–	–
Analysis sample	POI	226,305	Full BEC pharmaceutical file	47.2%
Winners-only price sample	Winning bid	196,883	Analysis sample	87.0%
Urgent panel	Urgent winning bid	56,803	Winners-only price sample	28.9%
Firm-buyer-item triple observations	Urgent winning bid	4,573	Urgent panel	8.1%
Firm-buyer-item triples	Triple	1,206	–	–
Both-regime urgent cells	Item-year-month cell	5,581	–	–
Singleton urgent cells	Item-year-month cell	11,972	–	–

Notes: Retention shares are reported only for nested samples with comparable units. The classifier universe is excluded from the denominator because it is a purchase-order/tender-notice file, whereas the rows shown here are POI-, winning-bid-, triple-, or cell-level empirical samples. POI denotes purchase-offer-item.

Table B.3: Sample means by purchase type.

	Ordinary	Administrative	Litigated
Observations (winners)	343,393	16,985	46,212
Log reference price	2.109	0.960	1.578
Log negotiated price	1.596	0.543	1.050
Quantity (units)	56,280	794,534	61,903
Number of bidding firms	3.159	2.394	2.191
Tender success rate	84.1%	85.7%	90.7%

Means computed on winners (POIs with an accepted winning bid). Reference and negotiated prices are in logs; quantity in units; tender success rate is the share of POIs that close with an accepted winning bid. Sample period: 2009–2019. Source: BEC-SP, SES/SP S-CODES.

Table B.4: Within-item balance between administrative and litigated urgent purchases.

Covariate	Mean admin	Mean litigated	Raw diff.	Within-item coef.	SE
SUS basic indicator	0.718	0.892	-0.174	—	—
Electronic auction	0.984	0.941	0.043	0.041**	0.017
Log quantity	7.642	6.180	1.462	0.866*	0.502
Log reference price	0.835	1.401	-0.566	-0.339***	0.093
Successful tender	0.869	0.914	-0.045	-0.024	0.022
Late period	0.565	0.623	-0.058	-0.045	0.041
Large PBU	0.895	0.810	0.085	0.083	0.073
Observations	63,426				
Items	2,370				

Notes: Within-item balance is restricted to urgent purchases for items observed in both administrative and litigated regimes. The within-item coefficient is from a regression of the covariate on the administrative indicator with item fixed effects and PBU-clustered standard errors. The SUS basic indicator is absorbed by item fixed effects in this specification and therefore has no estimable within-item coefficient. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.5: Representativeness of both-regime urgent cells.

Variable	Both-regime cells	Singleton cells	Difference / p -value
Log reference price	1.233	1.761	$p < 0.001$
Log negotiated price	0.703	1.326	$p < 0.001$
Log quantity	6.560	6.324	$p < 0.001$
Number of bidding firms	2.181	2.404	$p < 0.001$
SUS formulary share	86.7%	72.5%	$p < 0.001$
Reverse-auction share	95.5%	95.2%	$p = 0.039$

Notes: Both-regime cells are item-by-year-month cells containing both administrative and litigated urgent purchases. The table compares 5,581 both-regime cells with 11,972 singleton cells. The within-cell UTG specification is therefore identified from a selected, more formulary-intensive subset of urgent procurement.

C. Selection Bounds

Administrative requests are selected before purchase. Selection likely makes administrative cases easier or cheaper to fulfill than litigated cases, so naive administrative-versus-litigated comparisons can overstate the sanction-related price gap. The preferred partial-identification exercise trims the overrepresented administrative group within item $\times$ year $\times$ PBU strata.

All coefficients in this section are administrative minus litigated log negotiated prices. A negative coefficient means litigated purchases are more expensive. Percentage gaps are reported in the opposite, reader-facing direction: litigated-over-administrative prices. Top-tail and bottom-tail trimming of administrative observations provide the lower and upper Lee bounds reported in Table C.1.

Table C.1: Lee bounds on the under-the-gun price gap.

Specification	Admin coef.	SE	Gap (%)	N
Naive UTG: item + year + PBU FE	-0.259	0.092	29.5%	61,620
Lee lower bound: admin top-tail trim	-0.192	0.095	15.9%	45,624
Lee upper bound: admin bottom-tail trim	-0.148	0.097	21.1%	45,624

Notes: Coefficients are administrative minus litigated log negotiated prices. Negative coefficients mean litigated purchases are more expensive. Percentage gaps are reported as litigated-over-administrative price gaps. Trimming is applied within item $\times$ year $\times$ PBU strata where administrative observations exceed litigated observations. The mean trimming rate is 26.9%; the maximum trimming rate is 100.0%.

The trimming procedure affects 12,038 of 44,654 strata. The mean trimming rate is 26.9%, with maximum trimming rate 100.0%. The bounds are preferred to a parametric correction because the administrative channel is selected and the available exclusion does not provide credible stand-alone identification.

Table C.2 is included only to document why a parametric selection correction is not used as the main estimate. The second-stage correction is non-informative in this design; the paper relies on bounded selection plus mechanism evidence.

Table C.3 varies the trimming strata while holding the outcome model fixed. The exercise checks whether the Lee interval is driven by the preferred item-year-PBU trimming cells.

Table C.2: Parametric selection correction diagnostic.

Diagnostic quantity	Estimate	SE
Second-stage admin coefficient	0.817	4.278
Inverse Mills ratio coefficient	-0.653	2.637
Heckit ML ρ cross-check	1.000	—
First-stage pseudo- R^2	0.013	—
First-stage observations	5,351	—

Notes: This table is a diagnostic, not a preferred estimator. The exclusion variable is weak in this design, fixed effects absorb much of the useful identifying variation, and the resulting correction is not informative. The preferred appendix evidence is the Lee bounds, within-firm pricing tests, and sourcing measures.

Table C.3: Lee bounds under alternative trimming strata.

Strata	Admin coef.	Gap (%)	Trim (%)
item $\times$ year	[-0.326, -0.202]	[22.4, 38.6]	23.1
item $\times$ year $\times$ PBU	[-0.192, -0.148]	[15.9, 21.1]	26.9
item $\times$ year-month $\times$ PBU	[-0.202, -0.202]	[22.4, 22.4]	26.8

Notes: The outcome and second-stage fixed effects are held fixed across rows: log negotiated price with item, year, and PBU fixed effects. Rows vary only the strata used to compute the administrative trimming share. The item $\times$ year $\times$ PBU row is the preferred specification in the main analysis. Coefficients are administrative minus litigated log prices; percentage gaps are reported as litigated-over-administrative prices. In the finest item $\times$ year-month $\times$ PBU strata, cells contain too few administrative observations for top-tail and bottom-tail trimming to remove different observations, so the lower and upper bounds coincide; this row is reported for completeness, and the preferred bounds use the coarser item $\times$ year $\times$ PBU strata.

D. Additional Robustness

D.1. Supplier Fixed Effects and Quantity Controls

Table D.1 compares the urgent procurement price coefficient across specifications that add supplier fixed effects and log quantity. Attenuation after supplier fixed effects and quantity controls supports the interpretation that sourcing and scale, rather than a broad same-firm markup, account for much of the observed price margin.

Table D.1: Supplier Fixed Effects and Quantity Controls

	Baseline	Supplier FE	Supplier FE + quantity
Urgent purchase	0.051*** (0.015)	0.024 (0.016)	0.011 (0.018)
Log accepted quantity	—	—	-0.393*** (0.033)
Observations	196,883	196,642	196,642
R^2	0.868	0.881	0.922
Item FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
PBU FE	Yes	Yes	Yes
Supplier FE	No	Yes	Yes

Notes: The dependent variable is log negotiated price. The sample is accepted winning bids for items observed in both ordinary and litigated procurement. Column 1 includes item, year, and PBU fixed effects. Column 2 adds supplier fixed effects. Column 3 adds log accepted quantity. Standard errors, in parentheses, are clustered by PBU. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.2. Placebo on Never-Litigated Items

The placebo uses items never observed in litigation. It checks whether the urgent-procurement pattern appears for items outside the litigation margin.

Table D.2: Placebo Test on Never-Litigated Items

	Negotiated price		Reference price	
	Never-litigated	Main sample	Never-litigated	Main sample
Urgent purchase	-0.020 (0.032)	0.051*** (0.015)	-0.031 (0.045)	0.031** (0.014)
Observations	39,283	196,883	46,874	226,278
R^2	0.834	0.868	0.811	0.852
Item FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
PBU FE	Yes	Yes	Yes	Yes

Notes: The never-litigated sample contains items with zero litigated purchases during the sample period and variation between ordinary and administrative urgent purchases. The main sample contains items observed in both ordinary and litigated procurement. All specifications include item, year, and PBU fixed effects. Standard errors, in parentheses, are clustered by PBU. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3. Wild Cluster Bootstrap

Wild cluster inference addresses the finite number and uneven size of PBU clusters. Negative coefficients have the same sign convention as in Appendix C: administrative minus litigated log prices.

Table D.3: Wild cluster bootstrap inference on the UTG coefficient.

Specification	$\hat{\beta}$	Asy. SE	t	p_{boot}	95% CI (log-points)
Preferred (item + year + PBU)	-0.259	0.092	-2.80	0.008	$[-0.469, -0.039]$
Tightest (item $\times$ year-month + PBU)	-0.283	0.116	-2.44	0.039	$[-0.549, -0.001]$

Notes: Rademacher wild cluster bootstrap on the Admin coefficient with $B = 999$ replicates and restricted-null residuals; clustering at the PBU level. Negative coefficients mean litigated purchases are more expensive than administrative ones, in log-points. Source: `analysis/44_wild_bootstrap.R`.

D.4. Sample Restrictions

Table D.4 checks whether the under-the-gun price gap is driven by extreme quantities or items that appear rarely in urgent procurement. The estimates use the same sign convention as Appendix C.

Table D.4: Under-the-gun gap under sample restrictions.

Sample	Admin coef.	SE	Gap (%)	N	Items
Baseline UTG sample	-0.259	0.092	29.5	61,620	4,914
Exclude quantity tails	-0.214	0.078	23.9	60,993	4,883
Exclude rare urgent items	-0.270	0.102	31.0	53,202	1,355
Exclude quantity tails and rare items	-0.219	0.084	24.4	52,827	1,355

Notes: Each row estimates the same administrative-versus-litigated urgent price specification with item, year, and PBU fixed effects. The coefficient is administrative minus litigated log negotiated price; the percentage column reports the implied litigated-over-administrative price gap. “Exclude quantity tails” drops observations outside the central quantity range. “Exclude rare urgent items” keeps items with at least ten urgent winning observations. Standard errors are clustered by PBU.

D.5. Formulary Status and Market Depth

Table D.5 reports the under-the-gun price gap separately by formulary status and market depth. The table is a heterogeneity diagnostic for the deep-market interpretation, not a separate identification strategy.

Table D.5: Under-the-gun gap by formulary status and market depth.

Subsample	Admin coef.	SE	Gap (%)	<i>N</i>	Items
SUS-formulary items	-0.234	0.117	26.4	51,286	2,571
Non-formulary items	-0.098	0.098	10.3	10,334	2,343
High-depth markets	-0.356	0.178	42.7	7,975	1,007
Low-depth markets	-0.244	0.096	27.6	47,537	2,664

Notes: Each row estimates the administrative-versus-litigated urgent price specification with item, year, and PBU fixed effects within the listed subsample. The coefficient is administrative minus litigated log negotiated price; the percentage column reports the implied litigated-over-administrative price gap. Market depth uses the generated high-competition indicator when available, and otherwise splits observations by the median number of bidding firms. Standard errors are clustered by PBU.

D.6. Alternative Quantity Controls

Table D.6 checks whether the under-the-gun price gap is sensitive to the way accepted quantity enters the urgent price specification. The exercise is a scale-channel diagnostic, not a robustness check on the total under-the-gun effect: because quantity is itself a margin through which judicial urgency operates, conditioning tightly on log quantity removes part of the fragmentation channel studied in the paper. The BEC fields used here do not provide a common dose-standardized quantity measure.

Table D.6: Under-the-gun gap under alternative quantity controls.

Quantity control	Admin coef.	SE	Gap (%)	<i>N</i>
No quantity control	-0.259	0.092	29.5	61,620
Log accepted quantity	0.145	0.133	-13.5	61,620
Raw accepted quantity / 1,000	-0.243	0.086	27.5	61,620
Within-item quantity percentile	-0.199	0.066	22.0	61,620

Notes: Each row estimates the administrative-versus-litigated urgent price specification with item, year, and PBU fixed effects. The coefficient is administrative minus litigated log negotiated price; the percentage column reports the implied litigated-over-administrative price gap. Quantity controls are included as listed. The BEC data used here do not contain a common dose-standardized quantity field, so the table varies the observed accepted-quantity transformation instead. Standard errors are clustered by PBU.

D.7. Within Firm-Buyer-Item Heterogeneity

Table D.7 reports the same-firm, same-buyer, same-item test across subsamples. The administrative coefficient is administrative minus litigated. A positive coefficient means administrative purchases are more expensive within the triple; a negative coefficient means litigated purchases are more expensive within the triple. The deep-market null and the reappearance of positive coefficients in thinner or earlier subsamples are both part of the mechanism evidence.

Table D.7: Within firm-buyer-item null: robustness across subsamples.

Subsample	$\hat{\beta}_{\text{Admin}}$	SE	N
All triples (baseline)	0.035	0.041	4,573
Above-median quantity	-0.005	0.044	2,188
Below-median quantity	0.066***	0.025	2,027
SUS-formulary items	-0.001	0.026	2,540
Non-formulary items	0.101	0.064	2,033
Earlier period	0.117***	0.037	2,553
Later period	-0.038	0.052	1,492

Notes: Each row reports the within firm-buyer-item triple Admin coefficient on log negotiated price, with FBI and year fixed effects, PBU-clustered SEs. Coefficients are administrative minus litigated; negative values mean litigated purchases are more expensive within the same firm, buyer, and item, while values near zero indicate no detectable within-triple price difference. Subsamples are constructed from the triple sample (1,206 triples). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.8 keeps the baseline triple specification fixed and varies only the clustering level. This addresses whether the same-firm pricing result is an artifact of the baseline PBU-clustered variance estimator.

Table D.8: Within firm-buyer-item estimate: alternative clustering.

Clustering level	$\hat{\beta}_{\text{Admin}}$	SE	p -value	Clusters	N
Buyer/PBU (baseline)	0.035	0.041	0.401	20	4,573
Item	0.035	0.028	0.209	793	4,573
Item-buyer	0.035	0.026	0.192	1,099	4,573
Item + buyer/PBU	0.035	0.042	0.410	793; 20	4,573

Notes: Each row reports the same baseline within firm-buyer-item specification with FBI and year fixed effects. Rows vary only the clustering level used to compute standard errors. The coefficient is administrative minus litigated log negotiated price. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.8. Aggregation Within Buyer-Item-Month Cells

Table D.9 tests the direct aggregation margin. It compares administrative and litigated urgent purchases within common buyer-item-month cells, after collapsing each regime-cell to repeated purchase-offer-item demands and accepted quantity. The table is diagnostic evidence for the scale channel; it is separate from the within-firm triple design because aggregation can change the supplier set. The pattern is mixed and should be read with care: administrative cells carry larger total accepted quantity, consistent with greater scale, while litigated cells show more repeated purchase-

offer-items, consistent with more fragmented repeated demand. The exercise is therefore suggestive of fragmentation rather than a decisive aggregation estimate.

Table D.9: Aggregation of repeated urgent demands within buyer-item-month cells.

Metric	Administrative	Litigated	Admin – litigated	<i>p</i> -value
Purchase-offer-items per cell	1.01	1.11	-0.10	0.007
Cells with repeated POIs (%)	0.89	8.93	-8.04	0.006
Log total quantity per cell	5.76	4.96	0.80	0.004
Buyer-item-month cells	112			

Notes: The unit is a buyer×item×year-month cell observed with at least one administrative and one litigated urgent winning purchase. Each regime-cell is first collapsed to the number of distinct purchase-offer-items and total accepted quantity. The log-quantity row uses the log of total accepted quantity within the regime-cell. The table then reports paired means across common cells. *p*-values are from paired *t*-tests across cells. This diagnostic speaks to demand aggregation and does not use the firm-buyer-item triple sample.

E. Dynamic Evidence

The dynamic design is diagnostic and supportive; it is not the main identification strategy. The BJS event study assesses timing patterns around item-level exposure. Table E.1 reports the Honest-DiD sensitivity diagnostic. Because the first post-period estimate does not survive deviations at the observed pre-period maximum, the main claims rely on selection bounds, within-firm pricing tests, and sourcing evidence.

Table E.1: Dynamic-design sensitivity summary.

Diagnostic quantity	Value	Interpretation
BJS coefficient at first post period	0.052	First post-exposure timing estimate
Standard error	0.018	BJS period-specific standard error
Maximum absolute pre-period coefficient	0.041	Observed pre-period deviation scale
Breakdown linear-extrapolation <i>M</i>	0.018	Smallest linear-extrapolation allowance that reaches zero
Survives observed pre-period maximum	no	Diagnostic robustness indicator

Notes: The BJS event study is used to assess timing patterns, not as the primary identifying design. The final row records whether the first post-period estimate remains different from zero when the allowed post-period violation equals the observed maximum absolute pre-period coefficient. The sensitivity calculation uses the saved BJS period-specific standard errors; the off-diagonal covariance is unavailable in the source BJS event-study output.

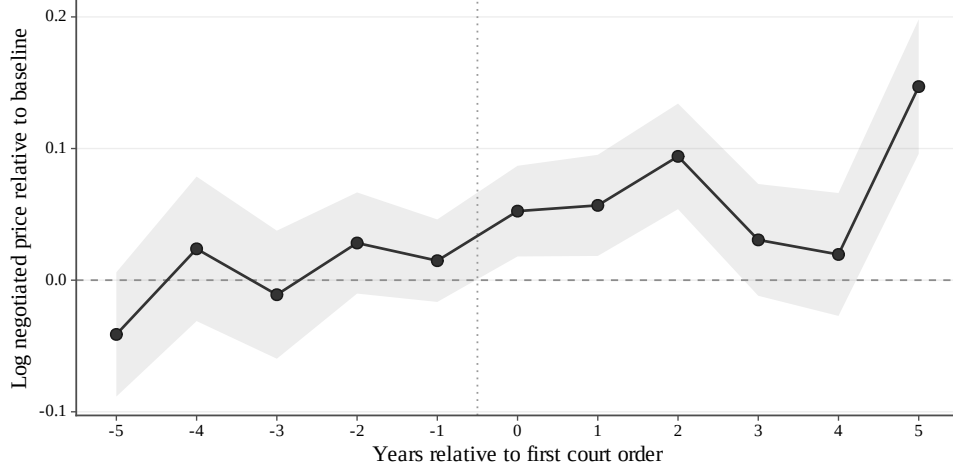


Figure E.1: BJS event-study diagnostic for item-level exposure.

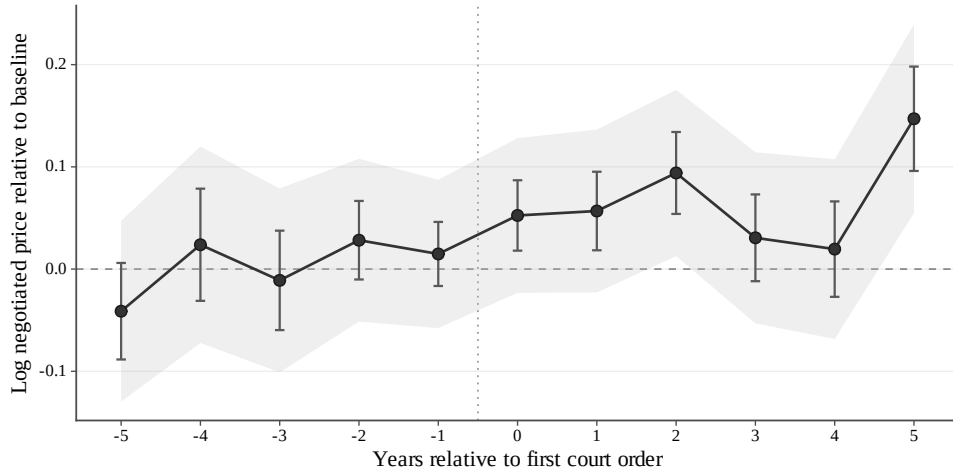


Figure E.2: Honest-DiD sensitivity diagnostic.

F. Procurement-Cost Calculation

The procurement-cost calculation applies the Lee midpoint 18.5% to an admissibility calibration of 50.0% and annual litigated spending of \$300 M. This yields \$27.8 M per year, with a Lee-bound range from \$23.9 M to \$31.7 M. This is a fiscal procurement-cost calculation, not a full welfare estimate.

Table F.3 reports how the fiscal procurement-cost implication changes under alternative admissibility calibrations. This sensitivity is mechanical: it varies only the admissible share of litigated spending and leaves the bounded under-the-gun gap unchanged.

Table F.4 instead varies the annual litigated-spending anchor while keeping the central admis-

Table F.1: Pricing-versus-sourcing reconciliation.

Component	Log-points	Admin-vs-litigated (%)	95% CI
Observed UTG gap	-0.259	-22.8%	[-35.2%, -9.1%]
Mechanical quantity component	-0.397	-32.8%	[-51.2%, -5.2%]
Within firm-buyer-item component	0.035	3.5%	[-6.6%, 11.0%]
Residual supplier-composition component	0.104	10.9%	[-16.0%, 39.6%]

Notes: Components are expressed in administrative-minus-litigated units. The decomposition uses 499 PBU-cluster bootstrap draws for confidence intervals. The residual composition component is a reconciliation residual and should be read together with the direct winner-switching evidence in the main paper.

Table F.2: Fiscal procurement-cost implication of the bounded UTG gap.

Component	Value
Lee midpoint	18.5%
Lee lower bound	15.9%
Lee upper bound	21.1%
Admissibility calibration	50.0%
Annual litigated spending	\$300 M
Annual procurement-cost implication, midpoint	\$27.8 M
Annual procurement-cost implication, lower	\$23.9 M
Annual procurement-cost implication, upper	\$31.7 M

Notes: This calculation is not a full welfare estimate. It applies the bounded price gap to a calibrated admissible share of litigated spending and excludes patient health benefits, search costs, and other welfare components. It should be read together with the sourcing decomposition in Table F.1.

sibility calibration fixed.

Table F.3: Sensitivity of the annual procurement-cost implication to the admissibility calibration.

Admissible share	Midpoint	Lee lower bound	Lee upper bound
25%	\$13.9 M / yr	\$11.9 M / yr	\$15.8 M / yr
50%	\$27.8 M / yr	\$23.9 M / yr	\$31.7 M / yr
75%	\$41.6 M / yr	\$35.8 M / yr	\$47.5 M / yr

Notes: Each row applies the bounded litigated-over-administrative price gap to annual litigated spending under the listed admissible share. The middle row is the calibration used in Table F.2. The table is a fiscal procurement-cost sensitivity calculation, not a full welfare estimate.

Table F.4: Sensitivity of the annual procurement-cost implication to litigated spending.

Annual litigated spend	Midpoint	Lee lower bound	Lee upper bound
\$250 M / yr	\$23.1 M / yr	\$19.9 M / yr	\$26.4 M / yr
\$300 M / yr	\$27.8 M / yr	\$23.9 M / yr	\$31.7 M / yr
\$350 M / yr	\$32.4 M / yr	\$27.8 M / yr	\$36.9 M / yr

Notes: Each row keeps the central admissibility calibration fixed at 50.0% and varies only annual litigated spending. The middle row is the spending anchor used in Table F.2. The table is a fiscal procurement-cost sensitivity calculation, not a full welfare estimate.